Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[10pt]beamer [utf8]inputenc [T1]fontenc graphicx
kpfonts fontspec Arial [onlymath]beamerfontthemeserif
amsmath mathabx graphicx siunitx commath xcolor tkz-euclide
braket cancel hyperref epigraph
empheq [most]tcolorbox [1][]nobeforeafter, math upper, tcbox raise
base, enhanced, colframe=yellow!65!black, colback=yellow!20,
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orangetext[2][enhanced, breakable, skin first=enhanced, skin
middle=enhanced, skin last=enhanced,
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boxed title to top center=yshift=-2mm, title=2,1**

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Title

Author Name[1]

Occasion, Location
19 Pebruari 2025

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ARLonghorn.png

[1] author_email@author_email_addres.com

Outline

Background

Development

Results

Conclusion

Objective It is good to identify the main objective(s) of the work. You can use colors to do so.

1. We will cover covers blah blah blah
2. Then we will see that blah blah blah blah

Background

A place for everything,
and everything in its place—Mother

- I like to include quotes as shown above using `epigraph`.
- Nonlinear terms can be added to right-hand side

[box=]equation* $\nabla^2 p - \frac{\partial^2 p}{\partial t^2} = \mathcal{O}(\epsilon^2)$

Background

[ams align, width=6.5cm, colback=blue!10,
colframe=blue!65!black] $\boldsymbol{\nabla}\cdot u + \frac{\partial \rho}{\partial t} = 0$
$\boldsymbol{\nabla}\mathrm{P} + \partial \rho u_{\overline{\partial t=0}}$

- Nonlinearities can be included by using the exact 5 and 5 equations
- Viscosity, heat conduction, and other lossy mechanisms not shown above

Development

- Parts of equations can also be highlighted: highlight math style=enhanced,colframe=red!90!black,colback=red!15,boxsep=0pt

$$[rememberas = fx]f(x) = \int_1^x \frac{1}{t^2}\, dt = \left[-\frac{1}{t}\right]_1^x$$

$$= -\frac{1}{x} + \frac{1}{1}$$

$$= [remember, overlay = [blue, verythick, - >$$

- What is the relationship between QM and acoustics?

[box=]equation* $i\hbar\frac{\partial}{\partial t}\Psi = \mathcal{H}\Psi$

Results

[size=fbox, colframe=orange!70!black,colback=orange!50!black,
fonttitle=,center title,drop fuzzy shadow]
[title=Lightning bolt]lightning.jpeg traffic_am.jpeg

Conclusion

- You learned this
- And you learned this
- But we didn't get to talk about this
- Next time we'll talk about this

Acknowledgements

- Funding source 1
- Funding source 2

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